

A1 Academy

Business Analyst Report - Sprint 1

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Project: A1 Academy - Group 06 - SE3022 Case Study Project

Sprint window: 19-30 August 2026

Epics covered: AA-1 (Initial Setup, Architecture & Deployment), AA-2 (Identity & Authentication Service)

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1. Introduction & Purpose

This report documents the Business Analyst work carried out during Sprint 1 of the A1 Academy project: requirements analysis, user story authorship, acceptance criteria, and backlog grooming for the Jira project AA. It complements the Sprint 1 QA Report by tracing each delivered story back to the requirement it satisfies and the tests that verify it.

A1 Academy is a centralised Learning Management System connecting Administrators, Teachers and Students through role-based access to one platform, replacing the fragmented, manual processes traditional tuition centres rely on for attendance, materials and communication.

2. Requirements Elicitation & Backlog Grooming

Requirements were drawn from the SE3022 case study brief and organised into two epics for Sprint 1: AA-1 (platform foundation - repository, database, event broker, CI/CD) and AA-2 (identity and authentication). Each backlog item was refined into a user story with a clear "As a / I want / so that" statement and acceptance criteria written as Given/When/Then scenarios, then estimated with the team on a 1-to-2-hour-per-story-point scale before being pulled into the sprint.

Definition of Ready

- The story has a clear As a / I want / so that statement.
- Acceptance criteria are written in Given/When/Then form.
- Dependencies on other stories or infrastructure are identified.
- The whole team agrees on the story-point estimate before sprint start.

Definition of Done

- All acceptance criteria are met by the implementation.
- Code is reviewed and merged into the target branch.
- Automated tests covering the story are passing in CI.
- Supporting evidence is captured under docs/evidence/.

3. Functional Requirements Coverage

Six functional requirement areas were identified for the full case study. Sprint 1 committed to the first of these in full; the remainder are scoped into later sprints against the epics they belong to.

Functional requirement	Sprint	Status
Secure JWT-based authentication & role-based access control	Sprint 1	Delivered
User moderation & subject category management	Sprint 2	Planned
Class scheduling & capacity-bound enrolment	Sprint 2	Planned
PDF material distribution & assignment submissions	Sprint 3	Planned
Attendance tracking, grading & gamification (digital badges)	Sprint 3	Planned
Dynamic analytics & progress reporting	Sprint 4	Planned

4. User Stories & Acceptance Criteria - Sprint 1

Five stories under Epic AA-2 made up the Sprint 1 feature scope. Each is reproduced below with the acceptance criteria used to define "done" and to brief the QA suite.

AA-12 Student Registration (3 SP)

As a prospective student, I want to create an account with my email and a password, so that I can access course materials and enrol in classes.

- Scenario 1 - Successful registration: Given a visitor submits a unique email, a valid password and their name, when the registration form is submitted, then a Student account is created with Active status and the user can log in immediately.
- Scenario 2 - Duplicate email rejected: Given an email address already exists in the system, when a visitor attempts to register with that email, then the registration is rejected with an "Email already exists" message and no duplicate account is created.
- Scenario 3 - Weak or missing password rejected: Given a visitor submits a password that does not meet the minimum security requirement, when the form is submitted, then the registration is rejected and the specific validation error is shown.

AA-13 Teacher Registration (2 SP)

As a prospective teacher, I want to register with my email, password and qualifications, so that an administrator can review and approve my account before I start teaching.

- Scenario 1 - Registration held for review: Given a visitor registers with the Teacher role, when the registration is submitted successfully, then the account is created with Pending status rather than Active.
- Scenario 2 - Pending teacher cannot log in: Given a Teacher account is Pending, when that user attempts to log in, then the login is rejected with the message "Your account is pending administrator approval."
- Scenario 3 - Approved teacher gains access: Given an Administrator approves a Pending Teacher, when that teacher next logs in, then access is granted and their dashboard loads normally.

AA-14 User Login (3 SP)

As a registered user, I want to log in with my email and password, so that I can access my role-specific dashboard.

- Scenario 1 - Valid credentials: Given a user with an Active account submits the correct email and password, when they submit the login form, then a JWT is issued and they are redirected to their dashboard.
- Scenario 2 - Invalid credentials: Given a user submits an incorrect password, when they submit the login form, then the login is rejected with a generic "invalid email or password" message that does not reveal which field was wrong.
- Scenario 3 - Blocked account states: Given a user's account is Pending, Rejected or Deactivated, when they attempt to log in with correct credentials, then the login is rejected with a status-specific message and no JWT is issued.

AA-15 Role Based Access Control (2 SP)

As an Administrator, I want each user role to see only the features relevant to that role, so that the platform stays secure and each user's interface stays uncluttered.

- Scenario 1 - Navigation reflects role: Given a user is logged in as a Student, Teacher or Admin, when they view the navigation menu, then only the links relevant to their role are shown.
- Scenario 2 - Server enforces role regardless of client state: Given a user attempts to call an API endpoint reserved for a different role, when the request reaches the server, then it is rejected with a 403 Forbidden response even if the client-side UI would have allowed the action.

AA-16 Password Reset (2 SP)

As a user who has forgotten my password, I want to reset it through a secure emailed link, so that I can regain access without contacting an administrator.

- Scenario 1 - Reset link issued: Given a user submits their registered email on the "Forgot password" form, when the request is processed, then a time-limited, single-use reset link is generated and sent to that email address.

- Scenario 2 - Reset completes successfully: Given a user opens a valid, unexpired reset link, when they submit a new password, then their password is updated and they can log in with the new credentials.
- Scenario 3 - Expired or reused link rejected: Given a reset link has already been used or has expired, when a user attempts to use it again, then the reset is rejected and a fresh link must be requested.

5. Non-Functional Requirements

Five non-functional requirement areas were defined for the platform. The table below states each requirement and how Sprint 1 addressed or verified it.

Requirement	How Sprint 1 addressed it
Performance	Login endpoint load-tested up to 200 concurrent users (AA-21).
Security	Password hashing and JWT claims verified at unit and integration level (AA-17, AA-18).
Availability	CI/CD pipeline and containerised deployment established this sprint (AA-11).
Usability	Registration and login forms validated at component level (AA-19).
Maintainability	Modular service boundaries defined (AA-9, AA-10); full test suite documented for AA-17 - AA-21.

6. Stakeholder & Target User Needs

User group	Core needs
Teachers	Class scheduling, digital materials, attendance, and engagement tools.
Students	Course catalogue, enrolment, submission tracking, and visible progress.
Administrators	Platform oversight, quality control, and security governance.

7. Traceability Matrix

Each Sprint 1 story is traced to the ticket that verified it, cross-referencing the QA Report's test layers.

Story	Verified by	Status
AA-12 Student Registration	AA-17 (unit), AA-19 (frontend), AA-20 (E2E)	Done
AA-13 Teacher Registration	AA-17, AA-18, AA-20	Done
AA-14 User Login	AA-17, AA-18, AA-21 (load)	Done
AA-15 Role Based Access Control	AA-18 (integration / role enforcement)	Done
AA-16 Password Reset	AA-17 (see QA Report, DEF-01: coverage gap noted)	Done

8. Sprint 1 Backlog Outcome

Metric	Result
Issues committed / completed	13 / 13
Story points committed / delivered	30 / 30
Completion rate	100% - zero carryover

9. BA Retrospective - Sprint 1

The team-wide retrospective is covered in the Sprint 1 Review; this section reflects specifically on the requirements and backlog-grooming process behind it.

What went well

- All 13 stories were pre-groomed with acceptance criteria ahead of planning, so every estimate was logged within five minutes of sprint start (4:17pm - 4:22pm, 19 August), with no re-estimation or scope changes for the rest of the sprint.
- Writing acceptance criteria in Given/When/Then form gave the QA suite a direct brief to test against, visible in the traceability matrix in Section 7.

What could improve

- Story sizes ranged from 1 to 5 story points without a shared reference scale, which contributed to the uneven workload split flagged in the Sprint 1 Review retrospective.
- Non-functional requirements (Section 5) were not written as their own backlog items, so their verification depended on being folded into existing stories rather than being tracked and estimated directly.

Actions for Sprint 2

- Agree a small set of reference stories (1, 2, 3, 5 SP) at planning so new estimates are anchored consistently.
- Give large non-functional concerns - performance, security - their own backlog items where they warrant dedicated verification, rather than relying on a related story to cover them.

10. Recommendations for Sprint 2

1. Begin grooming Epic 3 (Course Catalog & Category Service) and Epic 4 (Class Scheduling & Enrolment) in parallel with Sprint 1 close-out, rather than starting cold at the next planning session.
2. Keep every new story in Given/When/Then form so the traceability matrix in Section 7 stays consistent across sprints.
3. Introduce a short mid-sprint grooming check-in, echoing the Sprint 1 retrospective action item, to catch estimation drift before it affects delivery.
4. Pair each new access-controlled endpoint with its acceptance criteria for the 401/403 path at the time the story is written, closing the gap noted as DEF-02 in the QA Report.

11. Evidence & Reproducibility

- Jira Software project AA (A1-Academy), Sprint 1 board - tickets AA-9 through AA-21.
- Epics AA-1 (Initial Setup, Architecture & Deployment) and AA-2 (Identity & Authentication Service).
- docs/evidence/Teacher-Approval.md and docs/evidence/User-Deactivation.md - state-machine behaviour referenced in the AA-13 and AA-14 acceptance criteria above.
- Sprint 1 QA Report - test-level verification for every story listed in Section 7.

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